

Cloud Computing and Virtualization

Bachelors in Computer Engineering

**Prepared by Er. Ganga Gautam,
Assistant Professor, Pokhara University**

Chapter 1

Introduction(8 hrs)

Outlines:

1. Introduction of cloud computing
 - 1.1. Need for Cloud Computing
 - 1.2. NIST definition
 - 1.3. Characteristics and Benefits
 - 1.4. Application
2. Cloud reference model
 - 2.1. NIST Architecture
 - 2.2. Design principles of cloud architecture
 - 2.3. Infrastructural Constraints
3. Evolution of Cloud Computing
4. Cloud Service Model
 - 4.1. Software as a Service (SaaS), Platform as a Service (PaaS) and Infrastructure as a Service (IaaS)
5. Cloud Deployment Model
 - 5.1. Public, Private, Hybrid and Community
6. Jericho Cloud Cube model
7. Challenges and Ethical Issues

Cloud Computing

- Cloud computing is the delivery of computing services (servers, storage, databases, networking, software) over the internet.
- Cloud computing provides scalability, cost savings, and flexibility.
- Works through virtualization, resource sharing, and automation.
- Provides resources as a utility, just like electricity or water.
- Offers significant benefits but comes with security & reliability challenges.
- Enables anytime, anywhere access to data and applications.
- Managed by a cloud provider (AWS, Google, Microsoft, etc.).
- Netflix & other tech giants rely on the cloud for global service delivery.
 - Common Infrastructure
 - Location-independent
 - Online accessible
 - Utility that is available on
 - Demand

Need for Cloud Computing

- Scalability:
 - Dynamically scale resources as demand changes.
- Cost Reduction:
 - No upfront capital investment — pay per use.
- Performance & Reliability:
 - Cloud providers ensure uptime and maintenance.
- Mobility:
 - Access from any location or device.
- Focus on Core Business:
 - Offload IT management to the provider.

Real-World Examples

- Netflix:
 - Scales globally on AWS to stream to millions.
- Google Drive / Dropbox:
 - Cloud-based file storage.
- Salesforce:
 - Cloud CRM for businesses.
- Zoom / Microsoft Teams:
 - Cloud-based collaboration tools.

National Institute of Standards and Technology (NIST)

- U.S. federal agency promoting measurement and technology standards
- Provided a standardized definition of cloud computing (2011)
- Ensures common understanding across academia, government, and industry

NIST Definition(SP 800-145) of Cloud Computing

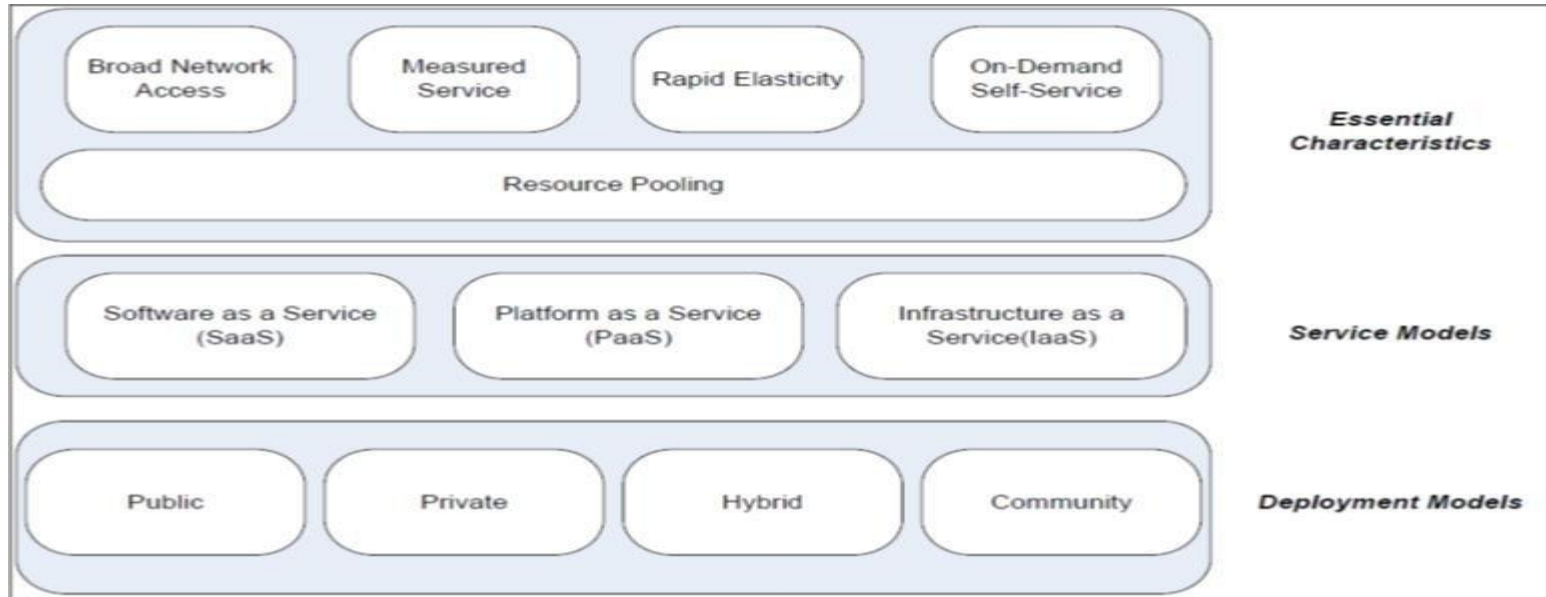
“Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources... that can be rapidly provisioned and released with minimal management effort or service provider interaction.”

Keywords:

- Ubiquitous access
- On-demand self-service
- Shared resources
- Rapid provisioning

NIST Cloud Model

- Five Essential Characteristics
- Three Service Models(SaaS, PaaS, IaaS)
- Four Deployment Models(Public, Private, Hybrid, Community)



NIST Essential Characteristics

1. On-demand self-service

- Users can provision resources automatically.
- Example:
 - AWS EC2 → self-provisioning

2. Broad network access

- Services available over the network via standard mechanisms.
- Example:
 - Google Cloud Storage → network-based access

3. Resource pooling

- Provider's resources are shared dynamically among users (multi-tenancy).
- Example:
 - Microsoft Azure → resource pooling

4. **Rapid elasticity**

- Scale resources up/down automatically with demand.
- Example:
 - Netflix scales during peak hours.

5. **Measured service**

- Resource usage is monitored, controlled, and billed per use.
- Example:
 - AWS bills hourly or per second.

Benefits of Cloud Computing

- Cost Efficiency:
 - Pay for what you use (OPEX model)
- Scalability:
 - Dynamically handle variable workloads
- Reliability:
 - High uptime and redundancy
- Flexibility:
 - Access anytime, anywhere
- Sustainability:
 - Better hardware utilization → greener IT

Applications of cloud computing

- Business
- Education
- Healthcare
- Government and public sector
- Entertainment and Media

Business Applications

- Customer Relationship Management (CRM):
 - Salesforce, HubSpot
- Enterprise Resource Planning (ERP):
 - SAP Cloud, Oracle Cloud ERP
- E-commerce:
 - Amazon, Alibaba Cloud
- Collaboration Tools:
 - Google Workspace, Microsoft 365
- Data Analytics:
 - AWS Redshift, Google BigQuery

Key Benefits:

- Cost reduction (CAPEX → OPEX)
- Centralized data access
- Rapid deployment and scalability

Educational Applications

- E-learning Platforms:
 - Coursera, edX, Google Classroom
- Online Labs & Virtual Classrooms:
 - AWS Educate, Azure Lab Services
- Research Collaboration:
 - Cloud-based data sharing and computation
- Resource Management:
 - University ERP systems on cloud

Benefits:

- 24×7 availability
- Accessibility for remote learners
- Cost-effective infrastructure

Healthcare Applications

- Electronic Health Records (EHR):
 - Cloud-based patient record systems
- Telemedicine:
 - Real-time consultation platforms
 - e.g., Practo, Telado
- Big Data Analytics:
 - Predict disease trends and patient outcomes
- AI in Diagnostics:
 - Cloud GPUs for medical image processing

Challenges:

- Security & privacy (HIPAA, GDPR compliance)
- Data sovereignty

Government & Public Sector

- E-Governance:
 - Citizen services, tax systems, digital ID
- Smart Cities:
 - IoT and data aggregation in the cloud
- Public Records:
 - Digital archiving and secure storage
- Disaster Recovery:
 - Cloud backup and resilience systems

Benefits:

- Transparency and accessibility
- Cost-effective scalability
- Enhanced data analytics for policy-making

Entertainment and Media

- Streaming Services:
 - Netflix, Spotify, YouTube (CDN + cloud hosting)
- Gaming:
 - Xbox Cloud, NVIDIA GeForce Now
- Content Delivery:
 - Cloud-based transcoding and storage
- Social Media:
 - Facebook, Instagram (massive distributed storage)

Key Technologies:

- Content Delivery Networks (CDNs)
- Elastic compute for variable workloads

Case Study – Netflix on AWS

- Netflix migrated entire infrastructure to AWS Cloud.
- Uses: EC2 (compute), S3 (storage), CloudFront (CDN), Lambda (automation).
- **Benefits:**
 - Global scalability
 - Reduced downtime
 - Faster content delivery
 - Cost efficiency

Netflix architecture overview diagram

Cloud Reference Model

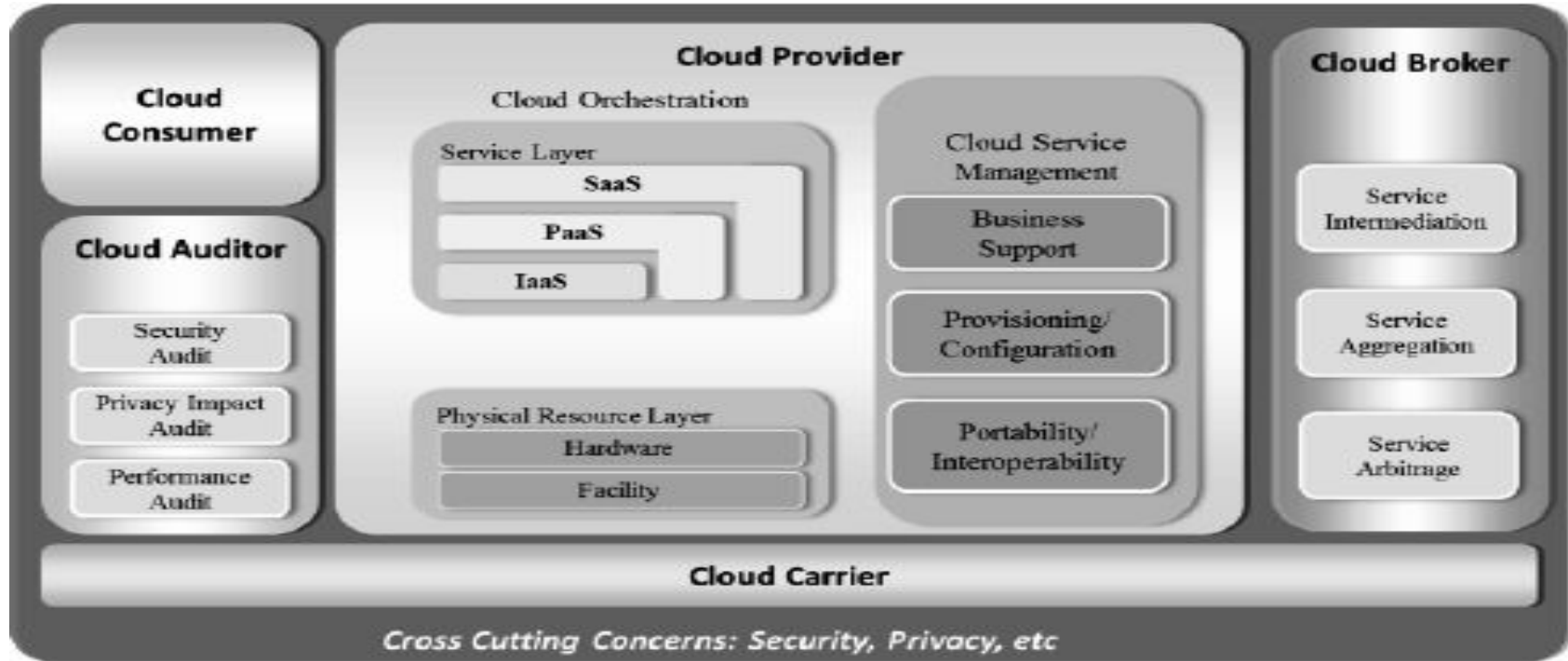
- Conceptual framework that describes the structure and relationships among cloud entities
- Provides a common vocabulary and set of abstraction layers
- Helps design, analyze, and compare cloud systems
- Foundation for interoperability and standardization

NIST Cloud Computing Reference Architecture (CCRA)

- Developed by NIST (National Institute of Standards and Technology)
- Defines roles, activities, and components for cloud systems
- Provides a structured framework defining the basic building blocks of cloud computing.
 - Serves as a reference point for designing, deploying, and managing cloud-based system
- **Goal:**
 - To ensure clarity and consistency in understanding and implementing cloud solutions.

For detail: <https://nvlpubs.nist.gov/nistpubs/Legacy/SP/nistspecialpublication500-292.pdf>

Layered View of Cloud Reference Model



Five major actors in NIST cloud computing reference architecture.

1. **Cloud Service Consumer**
 - end user or application using cloud services
 - Eg: End user using Gmail or AWS EC2
2. **Cloud Service Provider**
 - Offers cloud services (IaaS, PaaS, SaaS).
 - Responsible for infrastructure, configuration, and security (varies by service type).
 - Eg: AWS, Azure
3. **Cloud Broker**
 - Acts as an intermediary between consumer and provider.
 - Helps consumers select, integrate, or enhance cloud services.
 - Eg: cloud aggregator
4. **Cloud Carrier**
 - Transfers data between consumer and provider.
 - Functions like a network or internet service distributor.
 - Eg: ISP, telecom
5. **Cloud Auditor**
 - Conducts independent assessments of cloud services.
 - Ensures compliance, security, and performance standards.
 - Eg: Third party auditor

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, <i>Cloud Providers</i> .
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between <i>Cloud Providers</i> and <i>Cloud Consumers</i> .
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from <i>Cloud Providers</i> to <i>Cloud Consumers</i> .

Components of NIST Cloud Architecture

1. Service Models:

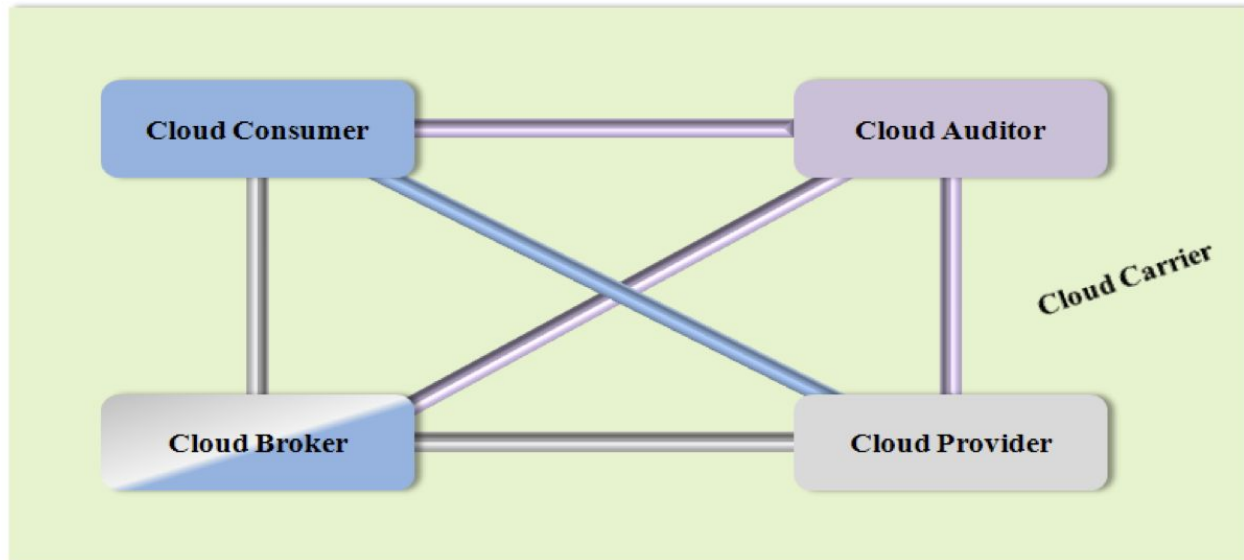
- SaaS:
 - consumer uses provider's applications on cloud
- PaaS:
 - consumer deploys applications onto provider's platform
- IaaS:
 - consumer provisions fundamental computing resources

2. Deployment Models:

- Public,
- Private,
- Hybrid,
- Community

Interaction among Actors

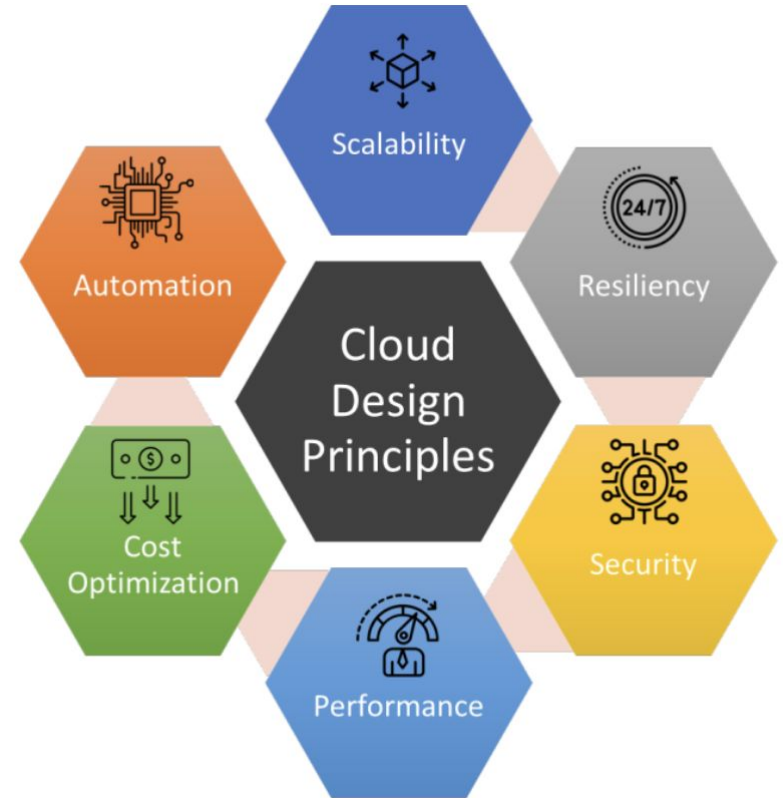
- Consumer ↔ Provider
 - service request and delivery
- Provider ↔ Broker
 - negotiation, optimization, and metering
- Provider ↔ Carrier
 - network transport
- Auditor ↔ All
 - independent assessment of operations and compliance
- Emphasis on trust boundaries and security domains



- The communication path between a cloud provider and a cloud consumer
- The communication paths for a cloud auditor to collect auditing information
- The communication paths for a cloud broker to provide service to a cloud consumer

Design Principles of Cloud Arc

- Foundation for reliable, scalable, and efficient s
- Key considerations:
 1. Scalability
 2. Resiliency
 3. Security and isolation
 4. Performance
 5. Cost Optimization
 6. Automation



Scalability and Elasticity

- Scalability:
 - Ability to handle growing workloads by adding resources
 - Horizontal scaling: Add more machines
 - Vertical scaling: Increase resources of existing machine
- Elasticity:
 - Dynamic provisioning/de-provisioning based on demand
- **Tools:**
 - Auto-scaling (AWS EC2)
 - Kubernetes

Resiliency

- Fault tolerance:
 - System continues operation despite failures
 - Ensure the system can withstand failures and recover gracefully.
 - Techniques:
 - Redundancy and replication(across availability zones/regions)
 - Load balancing/auto-scaling
 - Backup and disaster recovery
- Reliability metrics:
 - SLA uptime (e.g., 99.99%)

Security and Multi-Tenancy

- Multi-tenant model → shared infrastructure among multiple users
- Security Principles:
 - Isolation and sandboxing
 - Encryption (data-at-rest and in-transit)
 - Access control and identity management (IAM)
- Compliance: GDPR, HIPAA, ISO 27001

Performance

- Ensure the application runs efficiently and meets workload demands.
- Configure resources appropriately for performance and scalability.
- Choose suitable instance types, storage options, and networking setups based on workload patterns.

Cost Optimization

- Design the cloud environment to minimize resource wastage.
- Use reserved instances or spot instances when appropriate to reduce costs.
- Continuously monitor usage and spending to identify inefficiencies.
- Avoid over-provisioning; right-size resources according to actual demand.

Automated Design

- Automation:
 - Automated deployment (CI/CD pipelines)
 - Infrastructure as Code (Terraform, Ansible)
 - Auto-scaling and monitoring
- Use cloud tools and scripts so your cloud environment is consistent, repeatable, and efficient.

Infrastructural Constraints

Transparency

- Virtualization is key to sharing data and resources in the cloud.
- However, managing multiple servers and data sources makes complete transparency difficult.
- Proper resource visibility, load balancing, and usage reporting are essential to ensure scalability and performance.

Scalability

- Scaling cloud applications is complex — often requires reconfiguration or re-architecting systems.
- The infrastructure must support elastic provisioning (easy resource allocation and deallocation).
- Systems should be designed to scale dynamically according to demand.

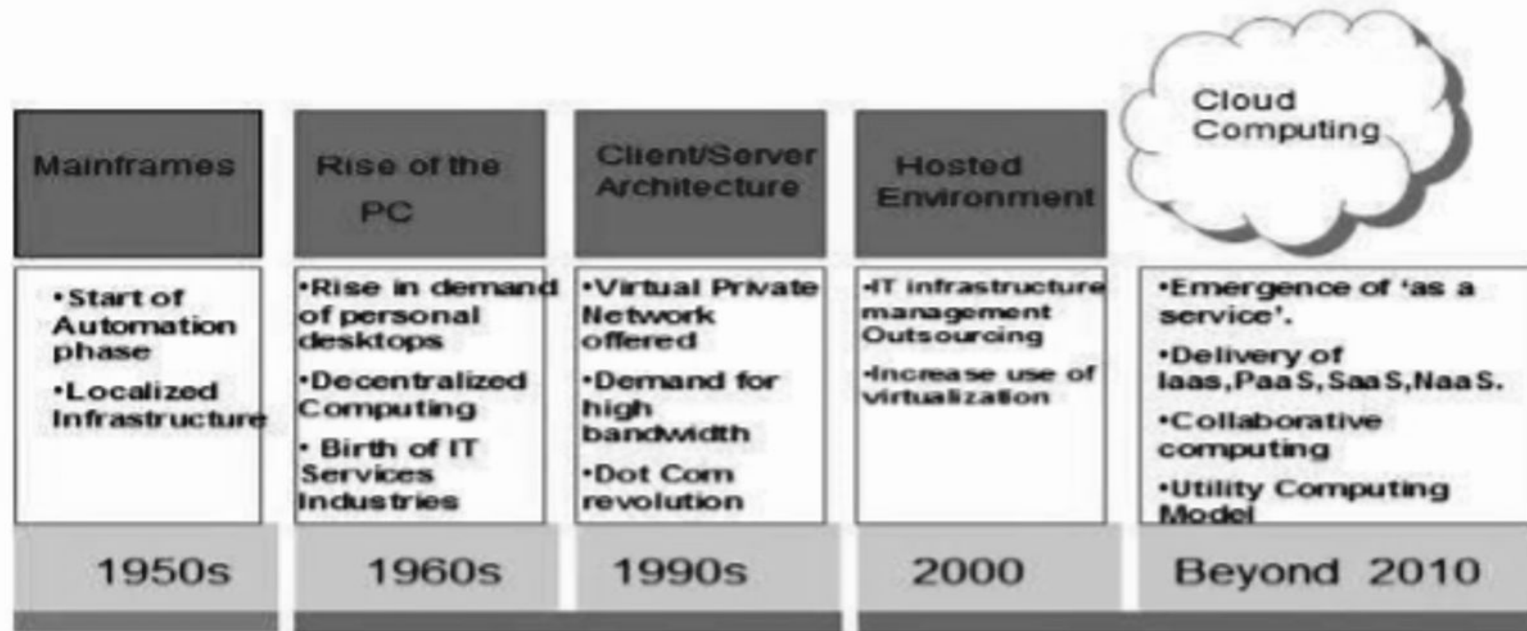
Smart Tracking

- To achieve transparency and scalability, intelligent monitoring tools are required.
- Smart tracking ensures continuous observation of performance, resource usage, and system health.
- Helps identify bottlenecks and optimize resource allocation in real time.

Security

- The data centers and access nodes (entry points) must be securely designed and protected.
- Security mechanisms should protect both stored and transmitted data.
- Critical to maintain data integrity, confidentiality, and safe access in large distributed environments

Evolution of Cloud Computing



1950s – Mainframes

- Beginning of the automation phase in computing.
- Systems were centralized and used localized infrastructure.
- Access through terminals connected to a single mainframe computer.

1960s – Rise of the Personal Computer (PC)

- Introduction and rising demand for personal desktops.
- Shift toward decentralized computing (individual users had local processing power).
- IT service industries began to emerge.

1990s – Client/Server Architecture

- Development of networked systems connecting clients and servers.
- Introduction of Virtual Private Networks (VPNs).
- Growing demand for bandwidth and the rise of the Dot-com revolution (web-based businesses).

2000s – Hosted Environment

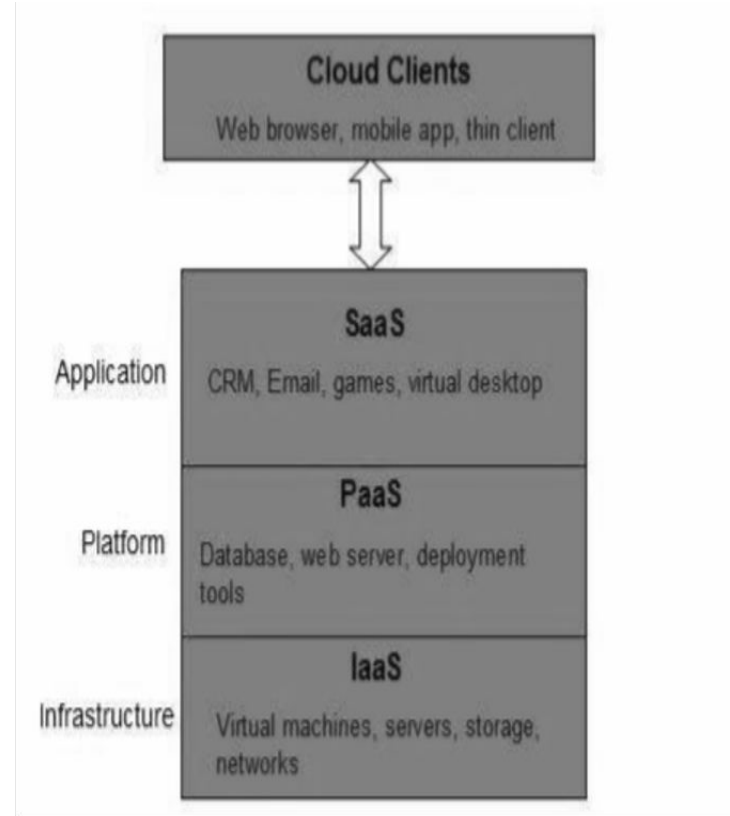
- Growth of IT infrastructure outsourcing.
- Widespread adoption of virtualization technology, enabling multiple virtual systems on one physical server.
- Transition from owning hardware to leasing IT infrastructure.

Beyond 2010 – Cloud Computing Era

- Emergence of the “as-a-service” model:
 - IaaS (Infrastructure as a Service)
 - PaaS (Platform as a Service)
 - SaaS (Software as a Service)
 - NaaS (Network as a Service)
- Rise of collaborative and utility computing — pay for what you use.
- Enabled scalability, on-demand access, and global connectivity through the internet.

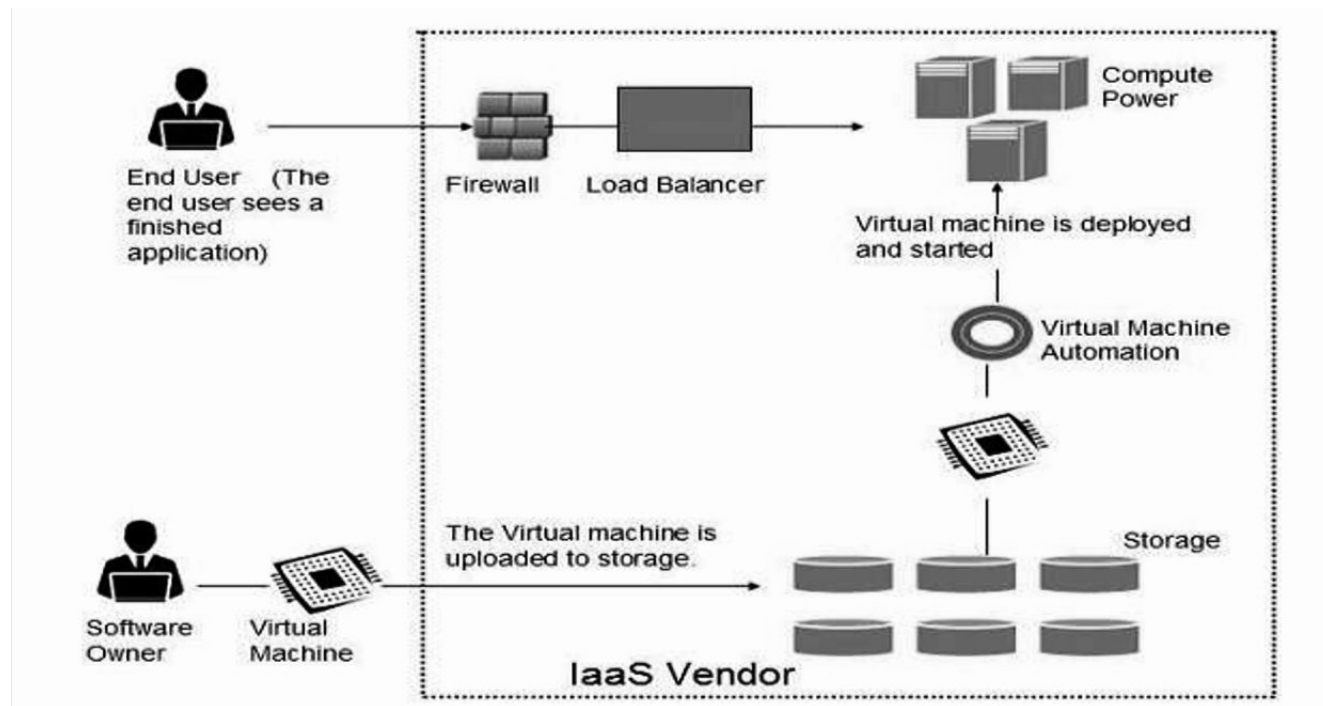
Cloud Service Models

- Cloud services are categorized into three main models, each serving different purposes:
 - SAAS
 - PAAS
 - IAAS



Infrastructure as a Service (IaaS)

- Provides virtualized computing resources like servers, storage, and networking to users through a virtual interface.
- Unlike PaaS and SaaS, IaaS doesn't include applications or even an operating system, it simply enables access to the infrastructure needed to support that software.
 - Provide extra storage for corporate data backups,
 - network bandwidth for a company website server, or
 - it can even enable access to high power computing which was previously only accessible to those with supercomputers.
- All of the above resources are made available to end user via server virtualization..
- Examples: AWS EC2, Microsoft Azure, Google Compute Engine
- Used:
 - Hosting websites & applications
 - Scalable storage solutions
 - Disaster recovery solutions

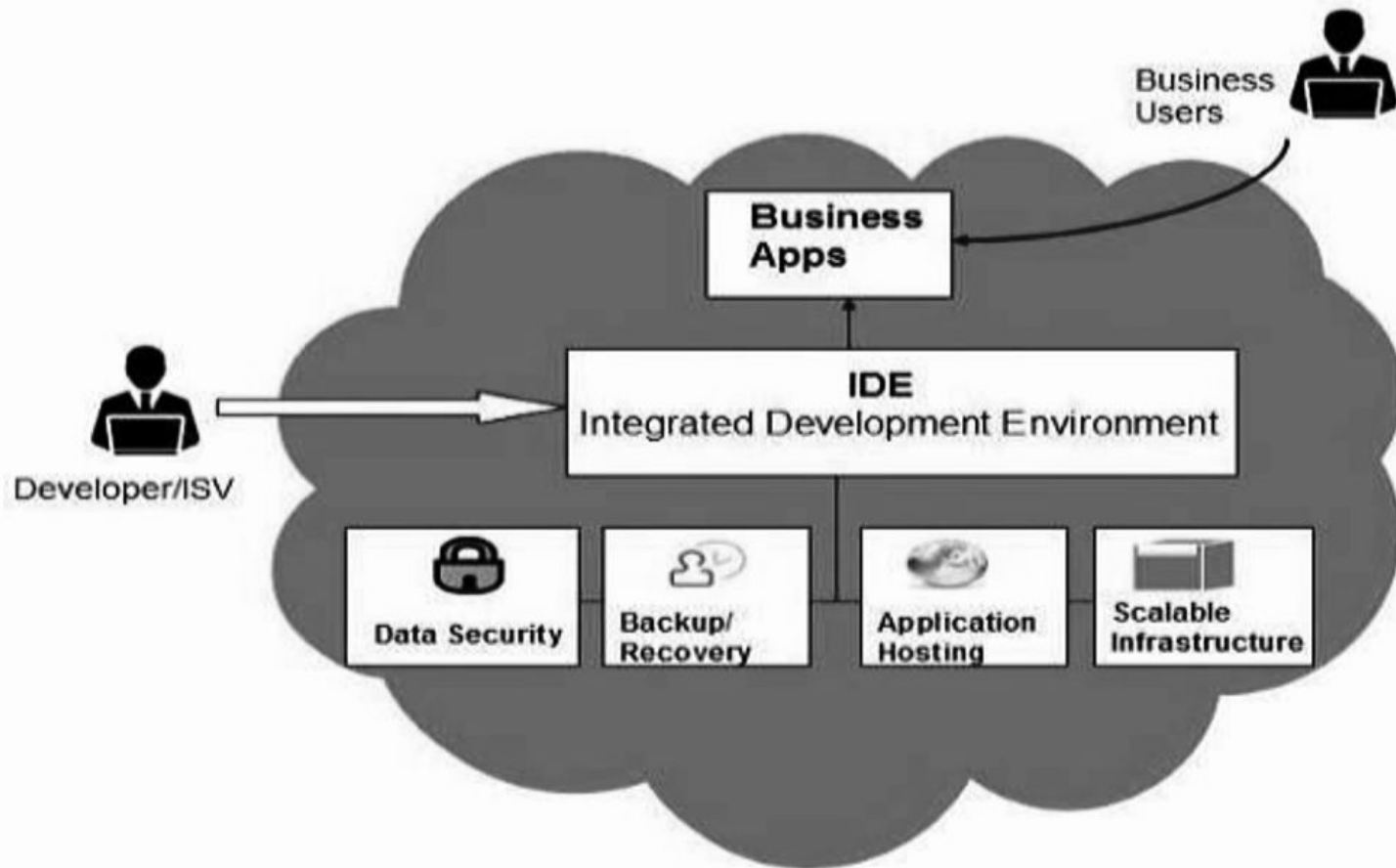


Benefits:

- IaaS allows the cloud provider to freely locate the infrastructure over the Internet in a cost effective manner.
- Some of the key benefits of IaaS are listed below:
 - Full control of the computing resources through administrative access to VMs.
 - Flexible and efficient renting of computer hardware.

Platform as a Service (PaaS)

- Provides a platform for developers to build, test, and deploy applications.
- cloud deliver a platform to users from which they can develop, initialize and manage applications.
- PaaS offerings typically include a base operating system and a suite of applications and development tools.
- PaaS eliminates the need for organizations to build and maintain the infrastructure traditionally used to develop applications.
- PaaS is sometimes called 'middleware', referring to how it conceptually sits somewhere between SaaS and IaaS.
- **Examples:** Google App Engine, AWS Elastic Beanstalk, Microsoft Azure App Services
- **Use Case:**
 - Developing & deploying web applications
 - Automating backend infrastructure
 - Simplifying database management



Benefits

- Lower administrative overhead:
 - Customer need not bother about the administration because it is the responsibility of cloud provider.
- Lower total cost of ownership:
 - Customer need not purchase expensive hardware, servers, power, and data storage.
- Scalable solutions:
 - It is very easy to scale the resources up or down automatically, based on their demand.
- More current system software:
 - It is the responsibility of the cloud provider to maintain software versions and patch installations.

Disadvantage : The developer locks-in with a particular vendor.

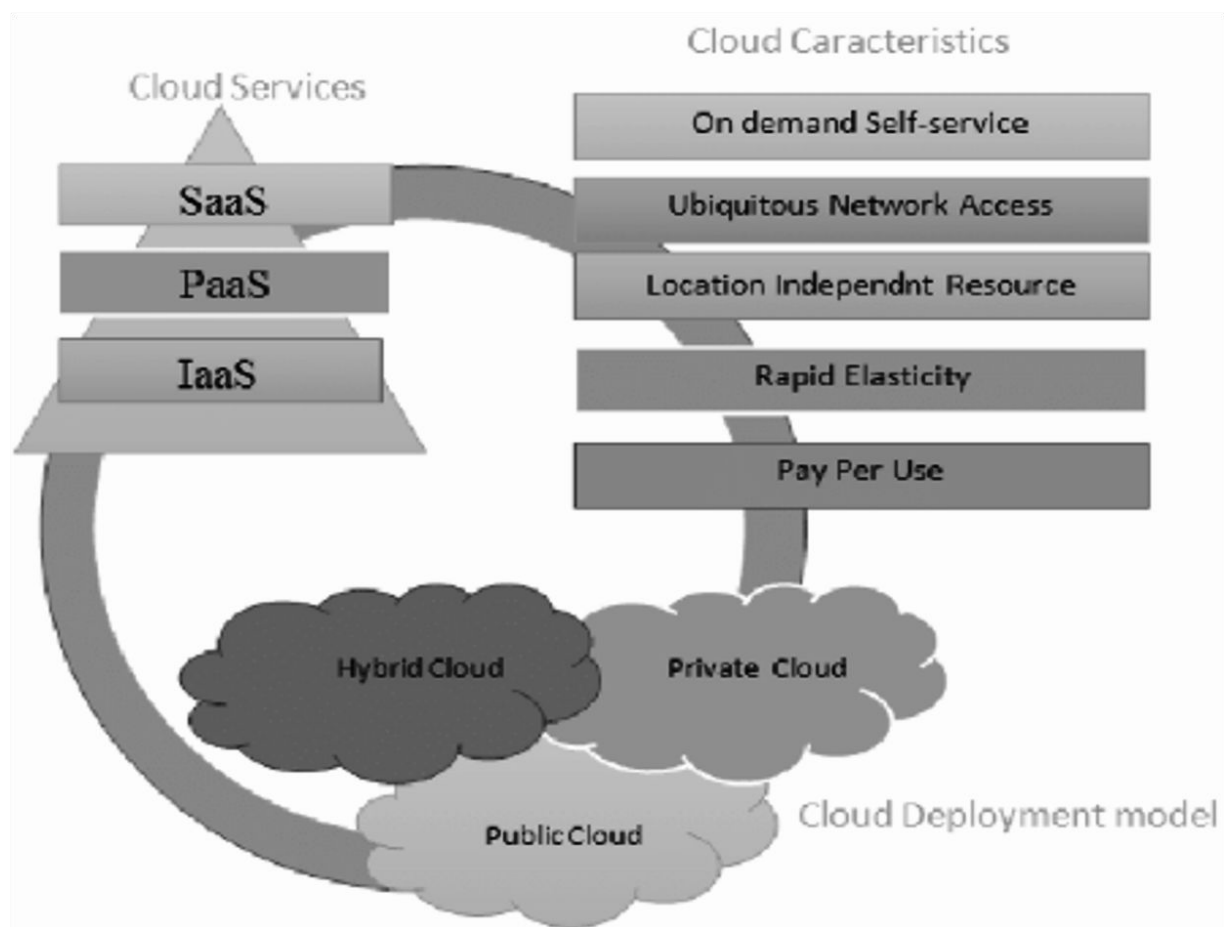
Software as a Service (SaaS)

- Provides ready-to-use software applications to the end users over the internet.
- Referred to as 'on-demand software',
- SaaS is a software licensing and delivery model
 - fully functional and complete software product
- SaaS offerings are typically accessed by end users through a web browser
 - can be billed based on consumption or, more simply, with a flat monthly charge.
- SaaS are most widely visible of all the cloud computing service models.
- **Examples:** Gmail, Dropbox, Microsoft 365, Salesforce
- **Use Case:**
 - Email & communication tools
 - Cloud-based storage solutions
 - CRM & business management apps



Benefits

- Low Cost:
 - Pay-as-you-go, no hardware needed.
- Automatic Updates:
 - Provider handles maintenance.
- Quick Deployment:
 - Ready to use instantly.
- Collaboration:
 - Multiple users work together.
- Reliable & Secure:
 - Backups, uptime, and data protection.



Comparison of Cloud Service Models:

Feature	IaaS	PaaS	SaaS
Control	High	Medium	Low
Flexibility	High	Moderate	Low
Maintenance	User-managed	Shared	Provider-managed
Examples	AWS EC2	Google App Engine	Gmail, Dropbox

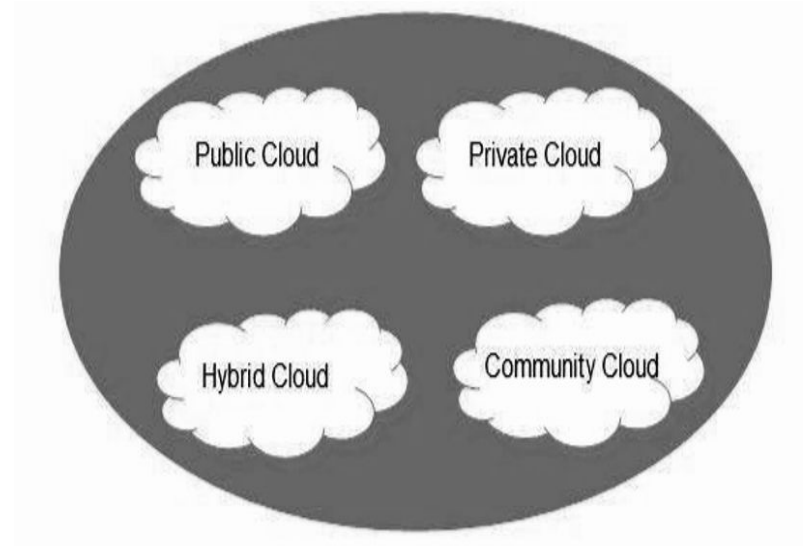
- IaaS, PaaS, and SaaS serve different needs, from infrastructure to software solutions.
- Choosing the right model depends on control, flexibility, and maintenance needs.
- Industries use a mix of cloud models for efficiency & scalability.
- Choosing the Right Model:
 - Startups → SaaS for easy adoption
 - Developers → PaaS for app building
 - Enterprises → IaaS for infrastructure control

Emerging Models:

- FaaS / Serverless – Function as a Service (e.g., AWS Lambda)
- XaaS – Everything as a Service

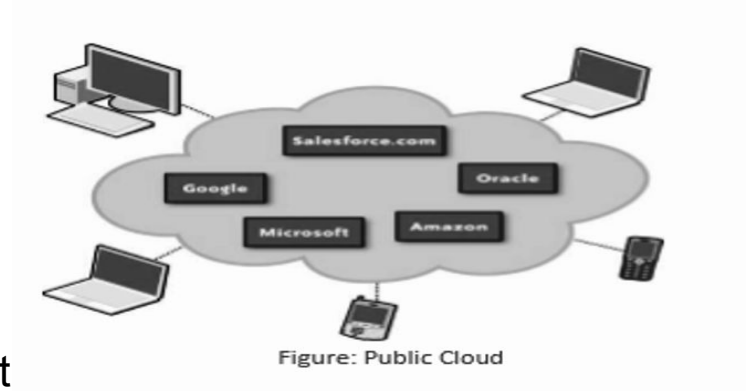
Cloud deployment model

- Public cloud
- Private cloud
- Hybrid cloud
- community cloud



Public cloud(External cloud)

- Services are accessible to the general public over t
- Operated and managed by a third-party vendor from one or more data centers.
- Follows the standard cloud computing model, offering applications, storage, and other resources.
- Low degree of customer control, as security and operations are handled by the vendor.
- Services may be free or based on a pay-per-usage model.



- Main Benefits of Public Cloud are:
 - Easy & Inexpensive Setup
 - Hardware, applications, and bandwidth are managed by the provider.
 - Scalability
 - Resources can scale to meet changing needs.
 - Cost Efficiency
 - Pay only for the resources you use; no wasted capacity

Private cloud(Internal cloud or corporate cloud)

- Proprietary computing environment providing services to a limited group of users behind a firewall.
- Leverages existing IT infrastructure, keeping core business operations in-house.
- Reduces maintenance burden once set up.
- IT teams act as internal service providers using virtualization and distributed computing.
- Accessible only within the organization and managed by internal IT.
- Ideal for critical and security-sensitive operations.
- Provides dynamic provisioning of computing resources.
- Internal billing can be used instead of pay-as-you-go.

Advantages of Private Cloud

1. Customer Information Protection
 - sensitive data remains secure within the organization.
2. Control and Customization
 - infrastructure tailored to organizational needs.
3. Compliance
 - easier to meet regulatory requirements.
4. Performance Optimization
 - resources dedicated to internal workloads.
5. Cost Predictability
 - predictable operational costs compared to public cloud usage.

Hybrid cloud

- Combines the best features of public, private, and community clouds.
- Allows companies to mix and match cloud types to suit their needs.
- Non-critical tasks run on the public cloud, while sensitive data stays on the private cloud.
- Hybrid cloud works best when data can be split into sensitive and non-sensitive categories.
- **Example:** Mission-critical workloads on secure private cloud, less sensitive workloads on public cloud.

- The benefits of a hybrid model are:
 - Improved Security & Privacy
 - Sensitive data stays protected.
 - Enhanced Scalability & Flexibility
 - Can handle varying workloads efficiently.
 - Cost Efficiency
 - Optimizes spending by using public cloud for non-sensitive tasks.

Community cloud

- Shared cloud setup for organizations with common goals and needs.
- Systems and services are accessible by a group of organizations.
- Similar to private cloud, but infrastructure is shared among multiple organizations.
- Best For: Joint ventures, research organizations, financial institutions, tenders.
- Shared infrastructure among trusted entities with uniform security, privacy, and performance requirements.
- Facilitates collaboration, project management, and implementation.
- Costs are shared across participating organizations.

Strengths

1. Cost Reduction
 - Shared infrastructure lowers overall expenses.
2. Improved Security, Privacy & Reliability
 - Maintains standards across members.
3. Ease of Data Sharing & Collaboration
 - Streamlined for joint projects.

Shortcomings

- Higher cost than public cloud.
- Fixed storage and bandwidth are shared.
- Limited adoption; not widely used yet.

Comparative Analysis

Feature	Public	Private	Hybrid	Community
Ownership	Provider	Single Org	Mixed	Multiple Orgs
Security	Moderate	High	High	High
Cost	Low	High	Medium	Shared
Control	Low	High	Medium	Shared
Example	AWS, Azure	VMware, OpenStack	Azure Arc	GovCloud

Jericho Cloud Cube Model

- The Cloud Cube Model helps categorize cloud deployments based on four dimensions.
- Developed by the Jericho Forum to assist organisations in choosing secure and appropriate cloud forms.
- Focuses on security, trust boundaries, ownership and control in cloud environments.
- The model includes four axes along which cloud deployments can be classified:
 1. Internal / External
 2. Proprietary / Open
 3. Perimeterized / De-Perimeterized
 4. Insourced / Outsourced

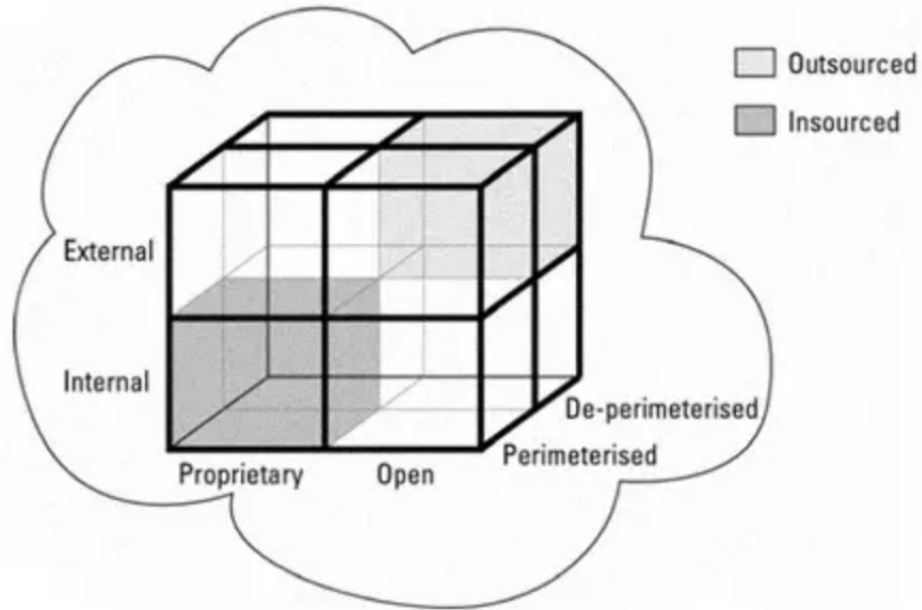


Fig:3D Cube showing classification
(Internal–External, Proprietary–Open, Perimeterised–De-perimeterised)

- **Internal / External**

- The location of data may be internally or externally which ultimately defines the organization's boundary.
- Internal : the cloud services or data reside within the organisation's boundary.
- External : the deployment/data is outside the organisation's boundary (e.g., a third-party cloud).

- **Proprietary / Open**

- Ownership is a measurement for not only ownership of technology but also its interoperability, use of data & ease of data-transfer & degree of vendor's application's lock-in.
- Proprietary: the technology, interfaces, ownership are restricted to one provider; limited interoperability.
- Open: the deployment uses more open standards or multiple suppliers, enabling easier data/partner collaboration.

- **Perimeterized / De-Perimeterized**

- Security Range measures whether the operations are inside or outside the security boundary, firewall, etc.
- Perimeterized: maintain traditional IT boundaries (e.g., firewalls, VPNs) around cloud resources.
- De-Perimeterized: beyond traditional boundaries: data may be encapsulated with metadata/security rather than relying purely on network perimeters.

- **Insourced / Outsourced**

- Sourcing defines whether the customer or the service provider provides the service.
- Insourced: cloud services are provided and managed by the organisation's own staff/infrastructure.
- Outsourced: cloud services are delivered via external third-party providers.

Example Classification using Jericho Cube

Cloud Type	Internal/External	Proprietary/Open	Perimeter	Management
Public Cloud (AWS)	External	Proprietary	De-perimeterised	Outsourced
Private Cloud	Internal	Proprietary	Perimeterised	Insourced
OpenStack Community Cloud	External	Open	De-perimeterised	Shared
Hybrid Cloud	Mix	Mix	Mix	Shared

Security Challenges in cloud computing

- Data Breaches:
 - Unauthorized access to stored data (e.g., misconfigured S3 buckets)
- Insider Threats:
 - Employees or contractors misusing credentials
- Data Loss:
 - Accidental deletion or hardware failure
- Insecure APIs:
 - Weak interfaces enabling attack vectors
- Virtualization Vulnerabilities:
 - VM escape, hypervisor attacks

Mitigation:

- Encryption, IAM, multi-factor authentication, monitoring

Privacy and Compliance

- Data Sovereignty: Data stored across countries with different laws
- Regulations:
 - GDPR (Europe) – right to privacy
 - HIPAA (USA) – healthcare data protection
 - PDPA (Asia-Pacific) – personal data rules
- Who owns the data? Who can access it?

Mitigation:

- Data anonymization, policy-based access, region-specific storage

Vendor Lock-In

- Difficulty moving data/apps between providers
- **Causes:**
 - Proprietary APIs, incompatible platforms
- **Effects:**
 - High switching cost
 - Reduced flexibility and innovation
- **Solutions:**
 - Open standards (e.g., Kubernetes, OpenStack)
 - Multi-cloud strategies

Interoperability and Portability

- **Interoperability:**
 - Ability of systems to work together
 - e.g., AWS ↔ Azure integration
- **Portability:**
 - Move applications/data without major modification
- **Challenges:**
 - Diverse APIs and storage formats
 - Differing VM/container images
 - Proprietary middleware

Solutions:

- Standard APIs (REST, OpenAPI)
- Containerization (Docker, OCI)
- Cloud management platforms (Terraform, Ansible)

Reliability and Performance Issues

- Downtime:
 - Even top providers (AWS, GCP) face outages
- Latency:
 - Data center distance affects user experience
- Resource Contention:
 - Shared infrastructure → noisy neighbor problem
- Disaster Recovery:
 - Requires redundancy & replication

Mitigation:

- Multi-region deployment, auto-scaling, failover strategies

Environmental and Cost Challenges

- **Environmental Impact:**
 - Massive data center energy use
 - Cooling and carbon footprint issues
- **Cost Management:**
 - Unpredictable billing (pay-per-use → bill shock)
 - Inefficient scaling wastes resources
- **Mitigation:**
 - Renewable-powered data centers
 - Monitoring tools (AWS Cost Explorer, Azure Advisor)

Ethical Issues in Cloud Computing

- **Data Ownership:**
 - Who truly owns user-generated data?
- **Transparency:**
 - Providers often hide operational details
- **Accountability:**
 - Responsibility during outages or breaches
- **Surveillance & Data Misuse:**
 - Unauthorized profiling or data resale
- **Ethical AI:**
 - Bias and discrimination when training on cloud data
- **Digital Divide:**
 - Unequal access to cloud technologies globally

Addressing Ethical Challenges

- **Principles for Ethical Cloud:**
 - Transparency in SLAs and data handling
 - Informed consent from users
 - Fair AI governance and data ethics
 - Corporate accountability and audits
 - Sustainable cloud practices
- **Global Initiatives:**
 - ISO/IEC 27018 – privacy for cloud services
 - Green Cloud Computing programs

Thank you